Diploma Project Proposal

MathVerity is a proposed platform for students in grades 5–6. Practice with restricted hints is followed by independent assessment and a parent report that explains the evidence behind each skill status.

Official topic: Development of an intelligent platform for improving school students' mathematical skills with gamification elements.

Working product name: MathVerity. The project concerns an intelligent platform for developing school students’ mathematical skills through practice and gamification. The official topic is retained; any refinement requires agreement with the supervisor.

Problem and Target Users

A student may obtain a correct answer using a hint or an external solver but still be unable to solve a similar problem independently. Parents may struggle to distinguish learning activity from skill mastery. MathVerity connects practice, restricted AI assistance, independent assessment and an explainable report. Families’ need for this combination remains a hypothesis to be tested through interviews.

The primary users are students in grades 5–6 who need additional practice, and their parents. A content author and a mathematics expert review items and rubrics; a separate school teacher dashboard is outside the MVP. Kazakhstan is the proposed pilot context, subject to confirmation. Curriculum alignment and content language must be agreed before items are released.

Aim and Research Questions

Develop and evaluate a web platform that supports selected mathematical skills through practice, pedagogically restricted hints and independent assessment, while giving parents verifiable evidence of progress.

RQ1: Does practice with restricted AI hints improve independent test performance compared with an otherwise identical practice tool using static hints? RQ2: How closely do the system’s skill statuses agree with expert assessment?

RQ3: Can parents correctly identify a difficult skill and the next step from the report? The contribution of gamification to motivation is investigated separately and is not treated as evidence of learning gains.

Objectives and Expected Outcomes

1. Analyse research approaches and existing products, and connect the findings to requirements.
2. Validate user needs through 3–5 discussions with representatives of the target audience, or obtain a substantive agreement with an actual client.
3. Design user roles, the item bank, independent assessment, the progress model and the parent report.
4. Develop a Go API, a React interface, PostgreSQL storage and a Docker environment.
5. Implement deterministic answer assessment and a restricted AI adapter with reviewed static hints as a fallback.
6. Prepare and conduct technical verification and a feasible educational pilot, and report the limitations of the results.

Expected deliverables include a reproducible prototype and MVP, an expert-reviewed item bank, documentation, test results, anonymised pilot aggregates and conclusions addressing the research questions. The platform does not promise to replace a tutor completely, verify the identity of the person answering or guarantee school admission.

Initial Version Scope

The planned MVP covers four topic blocks: ordinary fractions, decimals, percentages and simple linear equations. It includes eight skills (see Learning Content Model) and a target bank of 120 original items, with 15 per skill. This is a content plan, not an existing item bank. The scope is smaller than the full curriculum for grades 5–6 and makes it feasible to evaluate assessment and hint quality.

Included: a Russian-language interface; 10–15-minute learning sessions; numeric and fractional answers; diagnostic multiple-choice questions; explanations using structured steps; XP and achievements; secure parent linking; and skill reports. Kazakh may be added after translation and expert review.

Excluded from the MVP: full curriculum coverage through grade 6, handwriting OCR, video calls, payments, public child leaderboards, cameras and biometrics, cheating detection, training a proprietary LLM, deep knowledge tracing, school ERP integration and automated high-stakes assessment of children.

Participants and Current Status

Full-stack developer 1: Latiyev Abylay. Full-stack developer 2: Zhailanov Daniyar. Both developers share the same responsibilities: backend, frontend, integration and technical tests Supervisor: Yermaganbetova Madina. A mathematics expert reviews the educational content. Team membership and the allocation of responsibilities require agreement.

Analysis and design materials have been prepared. The production backend, full React client, actual AI calls, 120-item bank and educational experiment have not yet been implemented. Interviews and supervisor feedback remain to be included in the project materials.

Learning Content Model

This is the plan for the original MVP item bank. A teacher must review curriculum alignment and wording before publication. Full coverage of grades 5–6 is not claimed.

Table 1. Planned learning content

Code	Block	Example action	Typical error	Items
SK01	Fractions	Recognise equivalent fractions and simplify 6/8	Change only the numerator	15
SK02	Fractions	Compare 2/3 and 3/5 using a common denominator	Compare denominators alone	15
SK03	Fractions	Add 1/2 and 1/4	Add denominators	15
SK04	Decimals	Add 1.25 and 0.5	Misalign place values	15

Code	Block	Example action	Typical error	Items
SK05	Decimals	Multiply 1.2 by 0.3	Place the decimal point incorrectly	15
SK06	Percentages	Find 15% of 60	Confuse the proportion and the original amount	15
SK07	Percentages	Find what percentage 12 is of 60	Reverse the ratio	15
SK08	Equations	Find x in 3x = 21 and x + 7 = 19	Fail to use the inverse operation	15

Each skill is planned to have 8 practice items, 4 independent variants, 2 transfer variants and 1 delayed assessment item. Variants are generated from several mathematically reviewed templates to provide sufficient evidence; changing numbers alone does not create a new skill type. Independent variants are not shown in worked examples before the answer is recorded. Separate pre-test, post-test and delayed-test forms will be created for the pilot and are not part of these 120 items.

Prerequisites: SK01 → SK02 and SK03; SK04 → SK05; SK04 and basic division → SK06; SK06 → SK07. SK08 requires basic arithmetic. A missing prerequisite triggers a review recommendation rather than blocking the entire platform.

SK03 diagnostic example: for 1/2 + 1/4, the distractor 2/6 indicates a possible tendency to add numerators and denominators directly. A follow-up asks “Why is a common denominator needed?” and offers several explanations. Selecting 2/6 once does not establish a persistent misconception.

SK03 transfer example: “A strip is 1 metre long. Pieces of 1/3 m and another 1/6 m are cut off. What total length was cut off?” A subject expert checks that the item assesses the intended mathematical skill and that reading difficulty is appropriate for the age group. Fractional and decimal forms are accepted only when exactly equal.

Initial expert explanation rubric: 0 — no valid justification, or an incorrect operation; 1 — a partially stated correct method with a missing explanation; 2 — the transformation and its reason are stated correctly, and the answer agrees with the method. This rubric is intended for calibration between two experts, not automated LLM scoring.